

BRICS TRACK 2 • SUSTAINABILITY • INDIA-FIRST PROTOTYPE

AirSentinel

Locality-aware clean-air early warning and human authority review

Working prototype, evidence-first claims

Six-pollutant monitoring • 181.9-day locality evidence • 1h/3h/4h forecasts • sudden-spike review • citizen multimodal support • secure Google Cloud path

FINAL PROJECT REPORT

26 August 2026

Status: local implementation verified; cloud deployment requires team-owned credentials and billing.

Executive summary

AirSentinel closes the gap between city-wide air-quality visibility and a local government decision. It combines provenance-labelled outdoor observations, weather, contextual satellite layers and consented community evidence; forecasts one to four hours ahead; detects unsupported coverage or a sudden forecast miss; and creates a cause-unverified case for authenticated human review.

Main result

Najafgarh retains 7,516 PM2.5 rows and Safdarjung 7,889 rows across 181.9 days, with two reporting stations each. Their stored PM2.5 completeness is 86.1% and 90.4%, respectively. The source is labelled OpenAQ fallback data, not an authorised CPCB archive.

What judges can evaluate end to end

1. Inspect national/city conditions and locality evidence quality.
2. View a one-, three- or four-hour historical prototype forecast and uncertainty range.
3. Open an unexpected-spike or evidence-gap case whose cause remains unverified.
4. Submit privacy-minimised citizen evidence and keep it pending moderation.
5. Record a local demonstration review action, or use the private authorised API path for production-mode action testing.
6. Inspect honest Google integration readiness and secure deployment boundaries.

Truth boundary

- Not official AQI, a medical device, source attribution or automatic enforcement.
- OpenAQ is a labelled historical-development fallback, not presented as an authorised CPCB archive.
- No Google Cloud deployment is claimed until a team-owned project and verified URLs exist.
- Deep-learning and federated-learning models are not promoted in this submission.

1. Problem and solution

The four gaps

Gap	Why it matters	AirSentinel response
Scale	A city value can hide local events.	Separate locality anchors/geometries and evidence gates.
Coverage	Sparse or stale stations create false precision.	Show evidence gaps, station provenance, distance and completeness.
Forecast failure	A sudden event may lie outside the expected range.	Create a cause-unverified review case and seek corroboration.
Coordination	Station, weather, satellite and citizen data use different trust levels.	Canonical records, contextual labels, identity and audit.



CAUSE UNVERIFIED • UNCERTAINTY PRESERVED • NO AUTOMATIC ENFORCEMENT

CPCB current data when authorised | OpenAQ historical fallback visibly labelled | Context is not proof

Figure 1. Evidence is gated before forecasting; a sudden mismatch ends in human review, not automated blame.

2. Data improvement and trust

The historical collector now supports date chunks, pagination, retries, active-sensor probes, two working sensors per locality/pollutant, deduplication and safe merging. It preserves source-system labels so a coverage gate cannot silently upgrade a fallback into an official source.

Locality	PM2.5 rows	Span	Completeness	Stations	Pollutants
Najafgarh pilot area	7,516	181.9 days	86.1%	2	PM2.5, PM10, NO2, O3, SO2, CO
Safdarjung Enclave pilot area	7,889	181.9 days	90.4%	2	PM2.5, PM10, NO2, O3, SO2, CO

Evidence gate passed—not operational promotion

Both localities now support a controlled conventional baseline comparison. They still need approved geometry, station mapping, calibration/uptime context, weather alignment and fresh chronological evaluation before operational use.

Source hierarchy and alternatives

Priority	Source	Use	Limit
1	CPCB/data.gov.in current catalogue	Preferred official Indian current observations	Authorised key/configuration required; current data is not a guaranteed historical archive.
2	CPCB/SPCB/PCC approved export	Historical production evidence	Requires authority access, calibration and station metadata.
Fallback	OpenAQ	Historical development and discovery	Keep provider provenance and visible fallback label.

Priority	Source	Use	Limit
Context	Open-Meteo / Earth Engine / FIRMS	Weather, satellite gas/aerosol and fire context	Cannot independently prove surface PM2.5 or a source.

3. Forecasting and sudden-event handling

- Direct one-, three- and four-hour targets rather than recursively compounding a one-hour model.
- Leakage-safe lag, rolling, weather and available pollutant features.
- Time-ordered training/validation and newest untouched hold-out.
- Persistence first; established tree boosting only when it demonstrates stable improvement.
- Empirical residual ranges and an explicit research/prototype label.

What happens when the model is wrong

1. Detect that a fresh observation lies well outside the expected residual distribution.
2. Create an unexpected local spike case and preserve the observation/forecast evidence.
3. Mark the cause unverified; do not infer burning, industry, traffic or sensor failure.
4. Ask a reviewer to check nearby stations, sensor health, weather, satellite/fire context and moderated citizen signals.
5. Record the reviewer's monitor/inspect/escalate/close decision in the audit trail.

Model policy

The submission does not promote deep learning or federated learning. Vertex AI is only an optional host for a conventional, human-approved model that has earned promotion on approved data.

4. Working system components

Component	Job	Safety boundary
Public dashboard	Reviewed city/locality aggregates, coverage and forecast ranges	No personal evidence or enforcement actions
Citizen interface	Consented coarse-area text/photo/voice reports	Unverified pending evidence; exact location minimised
Authority portal	Case evidence and reviewer ownership; production UI is view-only	Writes use private authorised API; no automated external action
FastAPI	Typed contracts, readiness and integration endpoints	Fail-closed production authentication
Data pipeline	Discovery, collection, validation, provenance and coverage	Fallback labels cannot be promoted by volume
Forecast/anomaly	Short-horizon estimates and residual review cases	Historical prototype and cause-unverified language

5. Google technology architecture

PUBLIC AGGREGATES SEPARATED FROM PRIVATE EVIDENCE AND REVIEW ACTIONS



Figure 2. Public aggregates are separated from private evidence, identity and authority actions.

Technology	Meaningful job	Activation requirement
Gemini API / AI Studio	Guarded case explanation and in-memory photo description	Secret Manager value, feature review and output audit
Vertex AI	Optional conventional-model hosting/monitoring	Approved model, model card and monitoring plan
Firebase	Citizen identity and minimum pending-review metadata	Project, Auth, Firestore, rules, retention and abuse controls
BigQuery	Governed coverage/readiness/reviewed analytical tables	Dataset region, IAM, retention and cost policy
Maps / Earth Engine	Pilot-location and satellite contextual layers	Restricted key / registered project; contextual-only policy
STT / Translation / TTS	Accessible multilingual report and guidance workflow	Consent, language QA, retention and enabled APIs
Cloud Run	Public aggregate, private API and private authority services	Billing, IAM, secrets and URL/identity tests

6. Privacy, security and human governance

- Verified citizen identity in production; stronger reviewer identity for authority operations.
- Deny-by-default direct Firestore and Storage rules; server access through least-privilege cloud identities.
- No exact private address, raw media or unnecessary contact data in analytical tables.

- Bounded image/audio types and sizes; Gemini output cannot claim AQI, liability, enforcement or health diagnosis.
- Secrets stay in Secret Manager and are excluded from Git, Docker context and documentation.
- Production authority dashboard actions are view-only; writes must pass the private API's authenticated role policy and require durable storage before a pilot.
- The public Cloud Run image is built from a temporary six-file aggregate allowlist; private queues, moderation summaries and model features are excluded.
- Thirty-eight offline tests pass, including real in-process ASGI request-boundary checks and fail-closed cloud-adaptor tests.

Government-facing operating rule

AirSentinel may recommend what to verify next. An authorised officer decides whether to monitor, inspect, coordinate, issue guidance or close the case through the appropriate official process.

7. Revised phases and ownership

Phase	Delivered	External owner action
0 Foundation	Project, exclusions, evidence language and tests	Nominate data/privacy owner
1 Trusted data	Six-pollutant sources, contract and QA	Authorised CPCB history/current access
2 Long history	Najafgarh/Safdarjung 181.9-day fallback evidence	Verify mapping/calibration/geometry
3 Forecast & anomaly	1h/3h/4h research candidates and residual cases	Fresh approved-data re-evaluation
4 Authority review	Private queue, actions and audit	Official roles and escalation SLA
5 Citizen/accessibility	Text/photo/voice and moderation contracts	Firebase Auth, consent, retention, abuse response
6 Google services	Adapters and readiness checks	Project, APIs, secrets and quotas
7 Deployment	Docker and Cloud Run scripts	Run/test real deployment and identities
8 Submission	Reports, deck, script and checklist	Push GitHub, record video, insert URLs

8. Deployment and operational path

1. Create/select a team-owned GCP project, billing account, region and retention policy.
2. Authenticate gcloud and run the least-privilege bootstrap script.
3. Add reviewed secrets directly to Secret Manager and configure Firebase Auth/rules.
4. Create BigQuery governed tables and upload only approved analytical artifacts.
5. Deploy the private API and private authority portal from the private image; deploy the public dashboard from its temporary allowlisted context.

6. Test citizen identity, reviewer identity, audit persistence, negative access cases and all generated URLs.
7. Replace fallback/local file paths with approved persistent services before any operational pilot.

Current deployment truth

No GCP project, billing account, gcloud authentication or public URL exists in the local environment. The code and scripts are ready; the account-bound execution remains a team owner task.

9. Demo and submission package

Required item	Prepared artifact	Final human action
Source repository	Secret-safe source tree and release manifest	Create/push GitHub and add URL
3–5 minute video	Timed script and demo runbook	Record narration/screen and upload
10–12 slide deck	AirSentinel_Hackathon_Pitch_Deck.pptx	Add final URLs/team names if needed
Short description	SHORT_PROJECT_DESCRIPTION.md	Paste into submission form
Deployed link	Cloud Run scripts and status template	Deploy/test and add verified URL

Recommended demo story

1. Show the local monitoring gap and evidence quality.
2. Show Najafgarh/Safdarjung's improved historical coverage.
3. Show a short-horizon forecast and uncertainty.
4. Show a forecast miss becoming a cause-unverified case.
5. Show citizen evidence and a human authority action.
6. Show the Google service boundary and name the remaining account-bound steps honestly.

10. Limitations and next decisions

- Fallback history is not an approved government archive.
- Locality pilots are station-proxy areas until geometry and physical mapping are approved.
- Historical forecast performance is not live operational performance.
- Satellite vertical-column products are not surface PM2.5 in micrograms per cubic metre.
- Citizen, purifier and photo evidence supports triage but does not substitute for verified outdoor monitoring.
- Cloud Run local disk is ephemeral; operational reports/actions require Firestore/BigQuery persistence.
- Public warnings, health advice, source attribution and enforcement need the competent authority's policy and approval.

11. Source references

- CPCB/data.gov.in real-time air-quality catalogue — <https://www.data.gov.in/catalog/real-time-air-quality-index>

- OpenAQ API documentation — <https://docs.openaq.org/>
- Open-Meteo historical weather — <https://open-meteo.com/en/docs/historical-weather-api>
- Google Cloud Run authentication — <https://cloud.google.com/run/docs/authenticating/overview>
- Firebase security rules and authentication — <https://firebase.google.com/docs/rules/rules-and-auth>
- Gemini image understanding — <https://ai.google.dev/gemini-api/docs/image-understanding>
- Earth Engine Sentinel-5P catalogue — <https://developers.google.com/earth-engine/datasets/catalog/sentinel-5p>
- NASA FIRMS — <https://firms.modaps.eosdis.nasa.gov/>

Appendix A. Local commands

```
cd C:\Users\Rupesh\AirSentinel\AirSentinel
```

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -v
```

```
.\.venv\Scripts\python.exe -m streamlit run app.py
```

```
.\.venv\Scripts\python.exe -m streamlit run authority_app.py --server.port 8502
```

```
.\.venv\Scripts\python.exe -m streamlit run citizen_app.py --server.port 8503
```

```
.\.venv\Scripts\python.exe -m uvicorn src.api.main:app --port 8000
```

Appendix B. Submission message

AirSentinel

A locality-aware clean-air early-warning and human-review platform that combines provenance-labelled station data, weather, satellite context and consented community evidence to detect unsupported coverage and unexpected pollution changes, forecast one to four hours ahead, and guide an auditable sustainable response—without inventing hyper-local precision or automating blame.